

A survey of large language models for cyber threat detection

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ABSTRACT

With the increasing complexity of cyber threats and the expanding scope of cyberspace, there exist progressively more challenges in cyber threat detection and discovery. It's proven that some previous threat detection models may become inadequate due to the escalation of hacker attacks. However, recent research has shown that some of these problems can be effectively addressed by Large Language Models (LLMs) as an alternative to traditional methods. Nowadays, a growing number of security researchers are attempting to adopt LLMs for analyzing various cyber threats. According to investigation, we found that there is currently no systematic review of how LLMs are applied in cyber threat detection. To assist security researchers in learning and developing more solutions with LLMs efficiently, we have reviewed representative applications of LLMs on cyber threat detection. Firstly, we looked back brief history of LLMs and cyber threat detection, finding reasons why LLMs are suitable for some tasks of cyber threat detection. Next, we systematically retrieved and collated existing works in latest years, covering areas like threat intelligence, malware detection, threat prediction, etc. Additionally, We screened these articles according to certain criteria, then categorized typical works into common detection scenarios, assessing which step or point were LLMs optimized. Finally, we discussed extra topics including future directions and challenges in this field. The review aims to provide research status and comprehensive insights for researchers engaged in this field.

1. Introduction

Over the past several decades, the landscape of cyberspace threats has witnessed a progressive escalation in both frequency and complexity, culminating in widespread security incidents and substantial economic losses globally. This trend has unfolded in parallel with the evolution of tactics, techniques, procedures (TTPs), and strategies between hackers and security researchers (Kaur et al. 2023). Among various cyberspace threats, the Advanced Persistent Threat (APT) has emerged as a significant contributors to the security dilemmas faced by governments and corporations alike. For example, the APT organization FamousSparrow targeted key sectors across multiple Asian and African countries in 2021, causing serious chaos to telecommunications infrastructure. Besides, more and more similar security incidents have been continuously reported recently, such as Colonial Pipeline ransomware attacks in 2021, Kaseya VSA supply chain attacks in 2021, Microsoft Phonena malware in 2022, etc. Resulting of various APTs, the situation of cyberspace security came across an unprecedented crisis.

As threats become increasingly persistent, covert, and unpredictable, many traditional threat detection systems

struggle to identify the features of brand-new threats. Traditional methods, such as rule-based models, machine learning models, and deep learning models, are finding it increasingly difficult to adapt to the dynamic and evolving scenarios of threat detection and discovery (Yuan et al. 2021). This challenge primarily stems from the diversification of threat sources, the growing complexity of threat compositions, and the expanding scope of cyberspace. Meanwhile, cyber threats arise not only from vulnerabilities in information systems and hackers' activities but also from a range of other factors such as internal threats, espionage, security management errors, zero-day exploits, etc. Despite numerous efforts (Yaacoub et al. 2020) to enhance the level of automation and Intelligence in threat detection and discovery, security researchers continue to grapple with managing massive network data and intricate security tasks. Among the most pressing challenges are the limited scalability of detection systems, high false alarm rates in actual security scenarios, and vulnerability to adversarial attacks. Researchers increasingly recognize that addressing these challenges necessitates reassessing existing methodologies and seeking more robust, adaptable, and intelligent solutions in the realm of cyberspace threat detection.

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threats arise not only from vulnerabilities in information systems and hackers' activities but also from a range of other factors such as internal threats, espionage, security management errors, zero-day exploits, etc. Despite numerous efforts (Yaacoub et al. 2020) to enhance the level of automation and Intelligence in threat detection and discovery, security researchers continue to grapple with managing massive network data and intricate security tasks. Among the most pressing challenges are the limited scalability of detection systems, high false alarm rates in actual security scenarios, and vulnerability to adversarial attacks. Researchers increasingly recognize that addressing these challenges necessitates reassessing existing methodologies and seeking more robust, adaptable, and intelligent solutions in the realm of cyberspace threat detection.

Given this, some remarkable discoveries have suggested that Large Language Models (LLMs), particularly GPT and BERT developed in recent years, are capable of providing more effective solutions for some of these issues. Not only have LLMs shown superior performance in large-scale data works, multi-modal tasks content generation, etc., but can simultaneously support functions of perception, cognition, and decision-making in artificial intelligence (AI) tasks (Bommasani et al. 2021). As a result, there has been a surge in research utilizing LLMs for cyberspace threat detection and discovery, with these models rapidly gaining traction among researchers. Although numerous systematic reviews (Moustafa et al. 2019; Salim et al. 2023) have explored the main technological frameworks of threat detection, there is currently no systematic review comprehensively investigating the integration of LLMs with cyber threat detection. Therefore, we compiled this review to summarize some representative applications of LLMs in cyberspace threat detection and discovery, which may assist security researchers in efficiently navigating this emerging field.

The review is organized as follows: Section 2 delves into a brief history of LLMs and threat detection, finding interconnections between them. In Section 3, the majority of related works about applications of LLM on threat detection are presented in different security scenarios. In Section 4, we introduced the methodology followed for this review and further analyzed what we found. In Section 5, we shared extra opinions on related topics. In the end, we concluded this study and pointed out some future research directions and challenges in Section 6.

2. Background

The combination of AI and cyberspace threat detection dates back to the 1990s when security researchers attempted to apply Neural Networks (NNs) in Intrusion Detection Systems (IDS). From then on, an increasing number of researchers have employed AI methods to address security challenges. With the advancement of deep learning (DL), LLMs were born in the field of Natural Language Processing (NLP). After that, some applications of LLMs have emerged in cybersecurity, notably in threat detection (Gupta et al.

2023). In light of this, this section aims to outline some key milestones in the history of LLMs and cyberspace threat detection, shedding light on the inherent connections between these two domains.

2.1. Large Language Models

With the scale of NNs expanding, DL researchers were delighted to find that DL models have begun to exhibit a series of advanced abilities that traditional models can not demonstrate. These advanced abilities involve a chain of thought reasoning, human sentiment analysis, complex context abstracting, etc. This phenomenon is called the Intellectual Emergence (Zoph et al. 2022), which serves as an important indicator to distinguish Large models (LMs) and previous models. LLM, one of the LMs that is mainly applied in NLP, can effectively complete various language tasks such as text generation, question and answer (Q&A), intelligent retrieval, etc.

To achieve it, billions of parameters and efficient network structures like Transformer (Vaswani et al. 2017) are required. For instance, it was announced that there exist over 1.8 trillion parameters in GPT-4, one of the most worldwide popular LLMs. In addition, there are still numerous LLMs from other institutions like the open-source PaLM2, the superior LLM Gemini from Google, the advanced Llama from Meta, the Sora from Open AI, etc. Most of these LLMs have already achieved significant applications in tasks of finance, healthcare, education, customer service, etc. To realize LLMs better, it's necessary to introduce some momentous works in the history of LLMs.

When it talks about the origin of LLMs, can be traced back to the early NLP models. Mikolov et al. 2013 designed the Word2Vec method, compensating for the weakness of semantic mining in previous models. Through embedding words into a continuous vector space by shallow neural networks, the semantically similar words can be mapped to adjacent positions in the vector space. In this way, the model can predict other words based on context or target words, capturing more hidden semantic features. It was Word2Vec that first proposed the idea of text processing by unsupervised learning, which also laid the foundation for the emergence of LLM such as BERT, GPT, etc. Meanwhile, Cho et al. 2014 optimized Long-Short-Term-Memory (LSTM) networks, putting forward Gated Recurrent Unit (GRU) to improve the learning ability of language model for long-distance dependencies, which was also of great significance to model sequence for LLMs.

As time went by, more high-quality data and computing resources became available. AI researchers started to be concerned about transfer learning and Pre-Training models. As a result, there emerged two language models with epoch-making significance in 2018, namely BERT from Google and GPT from Open AI. Devlin et al. 2019 constructed BERT adopting a bidirectional Transformer architecture, which can simultaneously mine the context on both sides of word vectors. It was shown that BERT can learn universal language expressions and analyze contextual relevance

Table 1
The comparison of two prevailing style LLMs

Basic Architecture	Main Features	Representative Models
BERT - style (Encoder-Decoder or Encoder-Only)	Masked pretraining Rely on one-way context Judgmental model	ELMo, BERT, RoBERTa, DistilBERT, BioBERT, XLM, XInet, ALBERT, ELECTRA, T5, XLM-E, ST-MoE, AlexaTM
GPT - Style (Decoder-Only)	Auto-regressive pretraining Rely on two-way context Generative model	GPT-3, OPT, PaLM, BLOOM, GLM, MT-NLG, GLaM, Gopher, chinchilla, LaMDA, GPT-J, LLaMA, GPT-4, BloombergGP

through Pre-Training works on large-scale data, and fine-tuning on specific tasks. Consequently, BERT won excellent performances in multiple tasks such as expert Q&A, text classification, and entity recognition (ER). And Radford et al. 2018 chose the auto-regressive Transformer for GPT-1, with over 117 million parameters. Since GPT adopts a hybrid way of unsupervised Pre-Training and supervised fine-tuning, it can complete various complex NLP tasks, particularly text generation. In reviewing the history of LLM, can we find that BERT and GPT have jointly played milestone roles as well as represented two mainstream paths for latter LLMs, as shown in Table 1.

As mentioned above, BERT and GPT have set a widely accepted performance standard for new language models. Only two years later, Alec Radford et al. successfully developed GPT-2 and GPT-3. Notably, GPT-3 includes over 175 billion parameters and uses reinforcement learning with human feedback (RLHF) to extract transferable features. By this means, GPT has been competent in multitasking learning, inference, prediction, and zero-shot learning. Simultaneously, there has developed a series of improved BERTs such as RoBERTa (Zhuang et al. 2021), DeBERTa (He et al. 2020), applied in text processing, entity recognition, and intelligent translation, etc. However, people started to worry about various sectors of the security and ethical issues of LLMs at the same time. Over time, different LLM-driven products especially the ChatGPT have achieved technological and commercial success in different domains by 2023, pushing the LLMs to the forefront of AI capital. More and more LLMs are being developed, serving users with higher-quality content generation, multi-modal processing, deep semantic analysis, etc. Now, it has appeared a worldwide scene named the hundred-models War describing this phenomenon. Concurrently, there are more products like GPTs trying to help users build their own GPT requiring less or even no coding work. This work can greatly reduce the threshold for people to use LLM, but also catches our attention to AI security and privacy protection.

In view of these works, We can learn something about the development status of LLMs. Firstly, It is evident that DL models have witnessed a dramatic shift from quantity to quality, leading to an unexpected miracle. Although most of them still use the Transformer or other options as the model's fundamental unit, there exists a tendency of more parameters, more layers, and more input data in LLMs.

Besides, The open-source LLMs and closed-source LLMs can collaborate and promote each other. Many closed-source LLMs are optimized models based on open-source LLMs like Llama-2, but generate new ideas and data sets for optimizing open-source LLMs. With the development of LLM, the Encoders and Decoders are constantly being optimized, which is one of the most important upgrades. Most related products are concerned with innovation at an application level through the Pre-Training and fine-tuning paradigm.

Although LLMs have enjoyed a reputation in certain aspects, There are still some considerable limitations in LLMs. For example, we should learn that prompt injection, data poisoning attacks, backdoor attacks, etc. can sensibly decrease the quality of output and the model's efficiency. Meanwhile, model theft, model reconstruction, etc. will destroy the integrity and usability of LLM. Fortunately, most of such problems are being resolved by technical solutions or secure frameworks (El-Mhamdi et al. 2022). More and more users start to worry about their personal and sensitive information being leaked by LLMs. Therefore, some actions have been taken to cope with that and ensure output legal (Raeini 2023). Significantly, it is the excessive computational resource costs and environmental dependencies that prevent LLMs from being arranged on local systems at low cost. However, some methods of new gradient optimization and model simplification are proposed to solve this problem (Frantar et al. 2023), making LLM convenient enough for micro-enterprise or medium-sized enterprises.

2.2. Cyber Threat Detection

Since its emergence in the 1960s, the Internet has continually faced a myriad of threats such as viruses, vulnerabilities and Trojans. Simultaneously, the evolving nature of cyberspace encompassing digital technologies such as computers, communication networks, Internet of Things (IoT), etc., has correspondingly made cyberspace threats more complex and unpredictable. Generally, cyberspace threats (Aslan et al. 2023) refer to all events that have negative impacts on cyberspace security, ranging from natural disasters to management oversights and hackers' activities. The origins of threats include individual hackers, criminal organizations, and even other countries. Nearly all threats achieve it by exploiting technical vulnerabilities, system weaknesses, or human errors. According to their source, cyberspace threats

Table 2
Examples of external and internal threat

Threats Category	Examples
External threats	Malware, Phishing attack, Ransomware, Distributed Denial of Service (DDoS), Computer Viruses, Trojans, Zero-Day Exploits, Spyware, APTs, Network Intrusion, Exploit Kits, etc.
Internal threats	Users' Negligence (misconfigurations, unpatched systems), Malicious Users or Spies, Abuse of Privileges, Data Leakage, Spear Phishing, etc.

can be divided into internal threats and external threats, with typical examples shown in Table 2.

To address these threats, abundant threat detection and discovery works have been developed. The process of cyberspace threat detection (Al-Mhiqani et al. 2020) is comprehensive and changing, normally including the identification, evaluation, and prediction of various potential or known malicious entities or activities within cyberspace. To learn the current landscape in this field, we have systematically reviewed some representative-related works as follows.

In the early IDS or threats detection tasks, security researchers predominantly utilized Misuse Detection skills such as signature-based and rule-based methods, while also employing a limited use of Anomaly Detection skills like heuristic and statistics-based methods. On one hand, signature-based methods, relying on known threat characteristics, effectively identify threats by matching attributes of entities or behaviors against databases. The Rule-based detection achieves identifying potential threats by predefined rules, indicating attacks when network traffic exhibits certain abnormal patterns (Kumar et al. 2020). Conversely, heuristic analysis is more flexible, focusing on threat patterns rather than fixed characteristics, thus protecting against new and unknown threats. The statistics-based methods emphasize interpreting system state changes in networks and learning expected system behaviors to assess threat-containing traffic anomalies (Cao et al. 2017). Despite their efficacy in single-task detection scenarios, these methods are often trapped with high false positive rates(FPR), limited adaptability, dependence on known threats database, and challenges in effectively countering unknown threats.

With the advancement of AI applications, machine learning (ML) has driven the development of threat detection. ML for threat detection is primarily divided into supervised and unsupervised learning. On one hand, supervised learning focuses on classifying entities into threats or non-threats based on certain feature sets, such as differences between secure and threat data, and similarities among various threat data. There are plenty of examples of using ML in threat detection like Support Vector Machines (SVM), K-Nearest Neighbors (KNN), Naive Bayes Classifiers (NB),

and Logistic Regression (LR) (Almaleh et al. 2023). On the other hand, unsupervised learning emphasizes classifying objects into different categories based on structural features within the data, with related works like K-means, DBSCAN, Hierarchical Clustering, and Gaussian Mixture Models (Hou et al. 2022) for threat clustering. Although unsupervised learning does not rely on labeled datasets, reducing data dependency and human error, its outcomes require further analysis and are sensitive to noise and outliers. It is shown that ML is effective in analyzing superficial features, mitigating some inherent issues in early methods. However, ML may be inadequate when encountering large, complex datasets with in-depth threats. Moreover, ML usually requires extensive labeled data or domain expertise for training, posing practical application challenges in detecting new threats.

Owing to the complexities of threats characterized by non-linearity and high dimensionality, DL has been integrated into threat detection, leveraging different kinds of NNs and their variants to accommodate intricate feature engineering and tasks. The optimized Convolutional Neural Network (CNN) and Recurrent Neural Networks(RNN) have demonstrated efficacy in mining spatial and temporal features of threat (Ahmed Issa et al. 2023), thus enabling comprehensive network traffic analysis and continuous monitoring for APT. Besides, the Transformer, acclaimed for its proficiency in processing large volumes of sequential data, is well-suited for extracting threat content from diverse threat carriers (Xing et al. 2023). Likewise, the Graph Convolutional Network (GCN) has shown potential in analyzing the network's structure and nodes' relationships, especially in threat detection by provenance graphs (Ximing Li et al. 2023). Besides, Some variants of NNs like Generative Adversarial Networks (GANs) and Deep AutoEncoders (DAEs) have been proven to be efficient in the real-time discovery of new threats. Despite the unparalleled advantages of DL in identifying APT and dynamic threat analysis, there are still challenges, including high training and computing costs, limited interpretability of the model, and optimization problems, most of which remain largely unresolved to date.

Concurrently, some evolving technological resolutions are also put forward in this field. Firstly, Reinforcement Learning (RL), an AI paradigm predicated on learning and optimizing decisions through environmental feedback, plays a pivotal role in large-scale integrated threat detection systems (Yu 2022). Through refined reward functions, RL incessantly adjusts and optimizes detection strategies, thereby enhancing the online detection efficacy. In conjunction with other algorithms, RL contributes to the autonomous identification and mitigation of threats, facilitating threat hunting and adapting to new attack patterns significantly. In addition, visualization technologies and tools demonstrate extensive utility in analyzing large-scale events, traffic, logs, etc. Upon the visual models, computer vision (CV) algorithms can be used to analyze and identify threat entities, aiding security analysts in more intuitively learning and majoring security data and threat patterns (Nikolov et al. 2023). Moreover,

the development of threat detection based on Cyberspace Threat Intelligence (CTI) has reached new levels of maturity. From real-time and historical threat data, CTI can adeptly encapsulate advanced knowledge of diverse threats, aiding in cyberspace situational awareness. CTI can automate the identification of known threats and be used in the inference and prediction of unknown threats (Jiang et al. 2023). In all, These methodologies have significantly complemented previous methods, markedly curtailing the duration to discover attacks and amplifying the adaptability of threat detection systems. Nevertheless, each of them still possesses challenges needing further work, such as the complexity of reward functions designed in RL, the effectiveness of visualization tools on massive data, and the demands for data quality and timeliness in CTI, etc.

Last but not least, an array of cutting-edge technologies are increasingly coming to the fore. Initially, threat detection based on cloud computing and big data is being more widely implemented within cloud security products (Habeeb et al. 2019). Big data skills, such as data lakes and distributed computing, are employed for the efficient analysis of massive volumes of security data like logs, aiding security researchers in swiftly pinpointing and addressing threats. Subsequently, integrated methods amalgamating different threat detection approaches are emerging as a new trend in strategic solutions (Akram et al. 2023). These methods combine ML, DL, behavioral analysis, etc., to enhance the accuracy, comprehensiveness, and adaptability to real-world network environments, thereby reducing FPR and FNR. Apart from these, threat modeling and prediction are also providing robust support for the discovery of unknown threats (Zhao et al. 2022). Modeling in-depth information and analysis of attackers' behaviors and vulnerabilities, not only assist researchers in identifying real-time threats but also in forecasting potential threats and vulnerabilities effectively. Collectively, this avant-garde research is providing new resolutions for threat detection and discovery, promoting trends of threat detection towards high-level automation, high generalization, and cloud integration. However, these technologies are still in their nascent stages, facing various application challenges necessitating ongoing optimization.

Up to now, multiple threat detection and discovery methods have been introduced. Although there are still a few other types of these not mentioned, the studies summarized above are sufficient for us to learn research status in this field. It is shown that previous methods confront different challenges to some extent, especially in complex scenarios like cloud security. For instance, traditional threat detection and discovery methods often struggle with high FPR or FNR, indicating difficulties in discovering unknown threats, adapting to evolving threats, being misunderstood by constrained context, etc. These limitations hinder their effectiveness in the dynamic and data-rich landscape of cyberspace. Luckily, most of such problems are being solved by LLMs, renowned for their context-aware processing, comprehensive knowledge bases, and adaptive learning ability. Recent studies suggest that LLM could improve the accuracy, adaptability,

and generalization of threat detection in dynamic and data-intensive environments.

3. Thematic Works

In this Section, we will delve into multifaceted research utilizing LLMs for threat detection in specific scenarios. The reason why we did not categorize these works from other perspective (e.g input data types) is we aim to search the most closely related security scenarios to LLMs, which can help different researchers quickly learn the technological dynamics in their respective fields. Meanwhile, we found few literature indicating LLMs are directly applied to full life cycle of threat detection. On the contrast, most studies have shown LLMs were used for optimizing a certain process in existing research schemes. In light of this, the role of LLMs in different security scenarios may be more important than the processing objects of LLMs. The structure of this part will be carried out according to Figure 1.

As shown above, We have classified thematic works into four security scenarios that are closely related to the application of LLMs. The security scenarios involve correlative works of CTI, textual threat detection and social engineering attack prevention, malicious codes and malware detection, intrusion and threat discovery in comprehensive security scenarios. Since most of these works can be roughly divided into processes containing data collection and processing, threat feature extraction and analysis, target tasks like classification or clustering, etc., we can discuss which role LLMs played in different security scenarios step by step. Moreover, some evaluation criteria such as classification metrics (e.g. accuracy, precision, recall, F1 score), scalability, and robustness were used to estimate the performance improvements for previous threat detection models by LLMs. In this way, can this review effectively help researchers use LLMs for cyber threat detection in a targeted manner.

3.1. Applications of LLMs on CTI

To start with, we know LLMs were originally tailored for NLP tasks, and they can be used for entity recognition (ER) and relationship extraction (RE) on unstructured information, generating structured textual content. As a result, quite a few researchers have already applied LLMs to the production or application of CTI.

According to the definition by Gartner, CTI, intelligence strongly related to threat detection, refers to all evidence-based knowledge including mechanisms, labeling, actionable recommendations, meaning, context, etc., which is relevant to existing threats or potential threats faced by assets. Therefore, CTI has a high value on threat detection and response. In a narrow sense, CTI mainly includes trap identification used to identify and detect threats, such as file HASH, IP, domain name, program running path, registry key, and related attribution labels. Generally, The life cycle of CTI can be divided into six stages as follows: CTI targets, CTI collection, CTI processing, CTI analysis, CTI sharing, and CTI feedback. We found LLMs have been mostly applied in stages of CTI collection, processing, and analysis up to

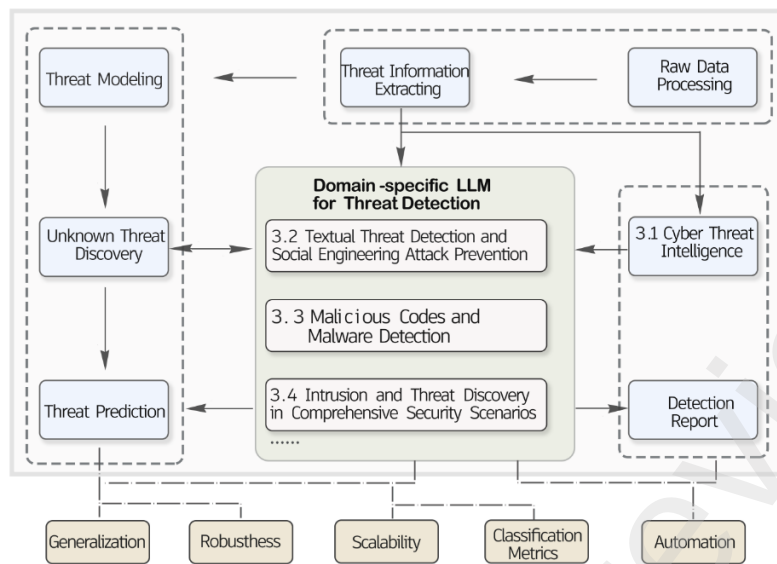


Figure 1: Framework of thematic works

now, as shown in Figure 2. In the remaining subsection of thematic works, we also use purple highlighted program blocks to indicate which part LLMs are involved in.

A. LLMs for CTI Collection

In most cases, the scarcity of high-quality data for CTI and the low generalization ability of detection methods make it difficult to identify unknown threats. To effectively collect CTI conforming specific network requirements, a large amount of labeled data is needed to train classifiers. Bayer et al. 2023 fine-tuned BERT and utilized data augmentation with text-davinci-002, combining three strategies of transfer learning, data augmentation, and few-shot learning to train high-quality classifiers from a few labeled instances. In the data augmentation phase, GPT-3 could generate new instances from a few existing labeled examples, significantly reducing the data labeling workload. With a limited number of data instances (testing with 32 instances), the classification results were comparable to those trained with 1800 instances, demonstrating the approach's effectiveness in assisting most of the data labeling work in CTI collection and threat detection. Besides, there exist a large amount of CTI data in open source data, mostly about IoCs (e.g. file hashes, IP addresses). Since this information is short-term effective and hackers can modify indicators of such information easily, most researchers prefer capturing data that can applied in threat analysis to collecting all of them. Jo et al. 2022 constructed the Vulcan system, which was able to extract various types of data describing CTI from unstructured text and determine the starting semantic relationships. According to its methodology, optimized BERT was used for RE and ER tasks for various threats. After training models, six types of CTI data and seven relationships between them could be recognized successfully. In test data, Vulcan could quickly analyze cyber threat entities and their relationships with 97 % and 98 % accuracy in ER and RE tasks, saving lots of time and cost for security researchers.

Furthermore, knowledge graphs can enhance the representation of knowledge, which is always used in CTI collection or processing. Ren et al. 2023 conducted a cyberspace security platform named CSKG4APT based on knowledge graphs. For unstructured text, CSKG4APT used CTI View based on BERT, constructed APT knowledge graph models by collecting and analyzing OSCTI, building fragmented information into APT scenarios containing 12 node types and 7 logical relationships. For updating the knowledge graph, an APT threat knowledge extraction algorithm is proposed, which utilizes DL and expert knowledge, distinguishing data associated with specific threats from irrelevant content. A framework showed attribution and countermeasures for APT. As it mentioned, CSKG4APT is not passive in security confrontation but can collect and integrate massive fragmented intelligence by LLMs to update security strategy. In addition, there are two main ways of presenting CTI, one is IoC, and the other one is unstructured reports. Zhou et al. 2022 optimized CTI View, which automatically extracted and analyzed CTI text published by various security vendors. To achieve it, a CTI extraction framework is designed based on automated testing, text recognition, and denoising. Through this framework, the adaptability of crawlers crawling heterogeneous CTI can be improved to a high level. In the process of threat entity extraction, a composite model named BERT-BiLSTM-GRU-CRF was declared to achieve better results on the labeled dataset than other existing models. IoC and TTP information described in CTI is also effectively extracted by using regular expressions combined with the black-and-white list mechanism.

B. LLMs for CTI Processing and Analysis

On CTI raw data obtained, some processing works need to be further executed. Wadhwa et al. 2023 assessed the performance of LLMs in RE tasks and found that GPT-3 with few-shot prompting, achieved near state-of-the-art (SOTA) performance. The Flan-T5 showed weaker capabilities in

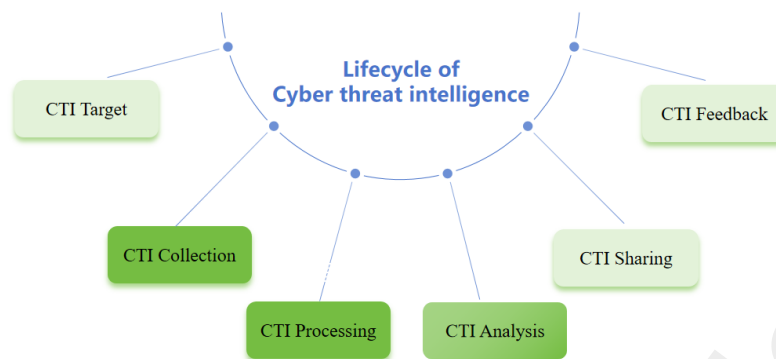


Figure 2: A diagram about LLMs in Three Key CTI Lifecycle Stages

few-shot settings, but supervision and fine-tuning through Chains of Thought (CoT) explanations generated by GPT-3 yielded SOTA results. It demonstrated LLMs can extract hidden and knowledge-level relationships from extensive security texts. Then, Sewak et al. 2023 developed CRUSH, utilizing GPT-4 and a semantic hypernetwork (LLM-EKG) to design and enrich Threat Intelligence Graphs (TIGs) for improved CTI processing and threat detection efficacy. In this model, LLMs effectively automated the extraction, inference, and validation of threat information from various sources, particularly excelling in identifying malicious scripts and mapping threat techniques. The CRUSH method achieved up to a 99 % recall rate in detecting malicious scripts and maintained over 90 % recall in identifying specific threat types. CRUSH could also standardize threat behaviors, enhancing the accuracy and efficiency of threat detection. Moreover, Vulnerabilities are an important component of CTI. A comprehensive vulnerability analysis can provide threat detection with important information such as vulnerabilities that attackers may exploit, potential threats, attacker techniques, and targets. Liu et al. 2023 used ChatGPT to map vulnerabilities to CWE-ID and ATT & CK technical IDs i.e. Vulnerability Description Mapping (VDM). Results showed that vulnerability description mapping to CWE-ID worked better while mapping to ATT&CK technology IDs worked worse. Unluckily, it seemed to be a serious problem in ChatGPT's conceptual understanding of CAPEC and ATT & CK technologies, and it is hypothesized that due to the high diversity of ATT & CK technologies and the poor quality of the data, the GPT is unable to understand the relevant technical content properly, resulting in a high rate of false positives. Therefore, GPT may be not the best option for VDM tasks.

C. LLMs for CTI Generation

It is always labor-intensive work to manually write security reports other than CTI. However, LLMs are competent in language analysis and generation capabilities, making them suitable for generating structured CTI. Perrina et al. 2023 proposed AGIR, an automated intelligence report generation tool using ChatGPT to generate STIX-standard CTI. Achieving a 99 % recall without introducing LLM

hallucinations, combining this information extraction system with AGIR can automatically process multiple reports, facilitating the automation of CTI tasks. And Siracusano et al. 2023 introduced a new large-scale open benchmark data set and aCTIon, a structured CTI information extraction tool. This data set comprises 204 real-world public reports and their corresponding structured CTI information in STIX format. As mentioned, researchers utilized GPT-3.5 to design prompts for two custom information extractions (ER and attack pattern recognition), then conducted self-checks based on the output results and validated results. Once verified, security reports in standardized STIX format are produced. In the same light, Gupta et al. 2023 argued that ChatGPT can automatically generate CTI based on a variety of data sources, including social media, news articles, dark web forums, and other online sources. By processing and analyzing large amounts of data, ChatGPT can generate reports that are accurate, comprehensive, and easy to understand. These reports help organizations identify potential security threats, assess their risk level, and take appropriate measures to mitigate them. And Mitra et al. 2024 proposed LOCAL-INTEL, which retrieves CTI from global threat repositories based on prompts and contextualizes them to a specific organization using its local knowledge database. LOCAL-INTEL consists of three key phases: global CTI retrieval, local knowledge retrieval, and contextualization completion generation. The former retrieves intelligence from the global threat repository, while the latter retrieves relevant knowledge from the local knowledge database. Finally, the fusion of these knowledge sources is orchestrated through a generator to produce contextualized completion results. The system automates the accurate customization of global threat intelligence against local data, helping Security Operations Center(SOC) analysts to reduce manual workload. From these works above, LLMs have demonstrated the ability to effectively deal with a large number of different knowledge sources. This capability is utilized to process global and local knowledge databases to automatically generate organization-specific CTI.

Except for CTI, some materials indicated that LLMs can be also applied in the production process of threat detection reports, situational security reports, or other textual

Table 3
Thematic works about LLMs on CTI

Tasks	Effectiveness	Limitations	literatures
CTI Collection	improving data labeling and data enhancement by GPT3.	Data enhancement and categorization worked bad when labels is too similar.	Bayer et al. 2023
	Scores for ER and RE tasks in security domain were 0.972 and 0.985.	Failure to validate the reliability of data and novel data beyond training.	Jo et al. 2022
	Effective integration of fragmented intelligence.	Inability to online knowledge completion and construction of knowledge, semantic gap between some IoCs.	Ren et al. 2023
	Using BERT to extract entities from bilingual CTI and update them into the APT Threat Knowledge Graph.	Inability to accurately locate entity boundaries, incomplete extraction of entities.	Zhou et al. 2022
CTI Processing and Analysis	GPT-3 achieves near state-of-the-art (SOTA) performance with few-shot prompting in ER.	Inability to perform severely on complex datasets; GPT3 without fine-tuned can be accessible only via interface.	Wadhwa et al. 2023
	LLM-EKG achieves 99 % recall on the task of detecting malicious scripts.	length for text analyzing is limited.	Sewak et al. 2023
	Vulnerability mapping to CWE works better than mapping to ATT & CK tech IDs.	ChatGPT can't be used directly for VDM, 'closed source is not the end of story'.	Liu et al. 2023
CTI Generation	Automated report generation with 99% recall rate.	Relying on ChatGPT may cause cost and privacy concerns.	Perrina et al. 2023
	Automated CTI extraction to generate structured STIX reports.	LLMs' hallucination problem in some tasks.	Siracusano et al. 2023
	Automated Report Generation.	hallucination problem in some tasks.	Gupta et al. 2023
	Automatically generate high-quality CTI to meet organization's needs.	Not mentioned	Mitra et al. 2024

cyberspace security reports, all of which are assistants for threat detection indirectly. From this part, it is obvious that LLMs have great potential in improving the automation of CTI production and the generalization ability of CTI aggregation, which can provide abundant informative support for threat detection-based CTI. All related works mentioned in this part are summarized in Table 3.

3.2. Textual Threat Detection and Social Engineering Prevention

As we know, Most input data for LLMs are natural language. so threat detection in textual content is one of the most closely related threat detection scenarios to LLMs. And Social engineering attacks typically rely on communication media that use text content such as messages, emails, public statements, etc. as carriers of information. Therefore, some researchers have applied LLMs to the threat detection of social engineering attacks, especially for phishing attacks, achieving satisfying results. Facing these threats, LLMs have a wide range of applications in textual content collection and processing, threat feature extraction and analysis, and textual threat classification, as shown in Figure 3.

A.LLMs for Threat Text Collection

In terms of data imbalance, LLM can generate extra raw data for training LLM or non-LLM detection models, thereby improving the overall generalization ability of threat detection. Jin et al. 2023 developed an efficient and scalable

Dark BERT, an LLM trained on a massive dataset of text from the dark web, to scan for emerging threats in the darknet. Dark BERT excelled in monitoring, investigating, and tracking illegal activities. It can also educate the public about the dark web, increasing their awareness of social engineering attacks. However, not all types of threat text are suitable for Dark BERT, it currently faces some challenges with bias, cost, and privacy protection which have not been fully resolved. And Sharma et al. 2023 experimented to study whether the phishing emails generated by humans and GPT can be identified, finding what are the differences in attack effectiveness. In over 40 practical tests by different groups, the attack effectiveness of manually crafted emails is only slightly higher than that of the GPT-3 email generation method. However, results may be restricted by the content, type, and implications caused by the experimental design of these emails. Simultaneously, Tsinganos et al. 2023 proposed SG-CSE BERT and a framework to identify the real intention of the interlocutor, which is essential for detecting social engineering attacks based on dialogue. With LLMs generating and analyzing historical information, they can output high-quality predicted attack states. And Shibli et al. 2024 presented a technical framework of phishing attacks based on prompt injection attacks by using AbuseGPT, and existing conversational models like Bard and ChatGPT. Creating smishing texts and real-world craftier smishing

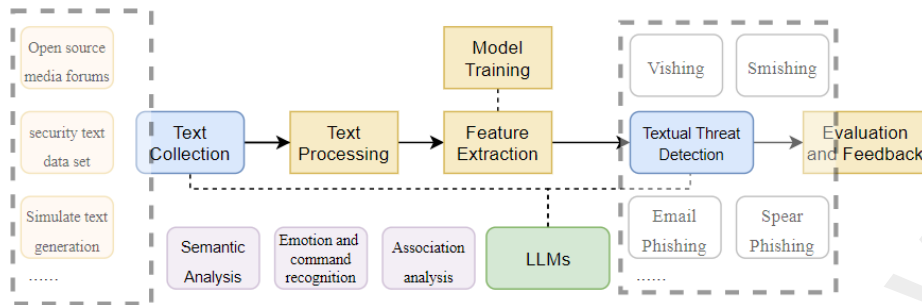


Figure 3: A diagram about LLMs' application in textual threat detection

campaigns has shown readers a vivid example of how to use LLMs to generate threat text.

Notably, Security Tools like FraudGPT and WormGPT can also generate abundant threat content that is not limited to malicious text. These weaponized generative AI tools can effectively assist hackers, such as malicious code/malware generation, phishing website creation, vulnerability analysis, etc. Since they are anonymous, illegal, and hard to find, few researchers chose to talk about them directly. But that does not mean we can ignore these tools because they may enable more attackers with the ability to carry out APT. Meanwhile, they have enormous application value in enriching threat corpora, indirectly enhancing threat detection capabilities.

B.LLMs for Threat Text Detection

In this section, some applications of LLMs or their APIs on threat detection for textual threat detection will be introduced. Ghourabi et al. 2023 combined Transformer and Ensemble learning with GPT-3 to detect spam and phishing messages, achieving a classification accuracy of 99.91 %. However, there are some limitations mentioned such as detection objects mainly being English messages, the poor scalability of models, and the small amount of test data. And Koide et al. 2023 conducted a system named Chat Phish Detector. It started with collected seed URLs to crawl for phishing and other websites, directly using ChatGPT to detect and achieve an accuracy rate of 98 %. Besides, this research declared LLMs performed well in detecting other social engineering threats. Unfortunately, detection results are unstable and susceptible to prompt injection attacks, and the training data has a deadline. But in terms of accuracy, the effect is good, especially for GPT-4. Moreover, Bilal et al. 2023 targeted hate speech issues on social media, developing a threat speech classification system based on BERT and Roman Urdu. Trained on a large dataset of 173,714 texts in a unique language, the study compared traditional machine learning and deep learning models, including LSTM, BiLSTM, BiLSTM with attention layer, and CNN, and explored transfer learning combining pre-trained BERT embeddings. The results showed that this transformer-based model surpassed traditional and deep learning models in accuracy (96.70%), precision (97.25%), recall (96.74%), and F-measure (97.89%), demonstrating stronger generalizability in cross-domain datasets. Kumarasiri et al. 2023 adopted MobileBERT, DistilBERT, and TF-BERT to complete tasks

such as message content analysis, URL category analysis, HTML content analysis, etc. This work can help mobile devices identify phishing information more accurately. Nahmias et al. 2024 focused on better feature representation for phishing emails, and designed a problem list to let LLMs infer the different attributes of the mail. After that, a feature representation was constructed for the downstream machine learning classifier. The test results indicated that the accuracy of the detection exceeds 87% and F1-score exceeds 91%. This method has good generalization ability and it can be further optimized through model tuning or case analysis of misclassified samples.

Furthermore, some other reports may declare LLMs can also be applied in internal threat detection, social forum speech monitoring, sentiment analysis of threats, and intention recognition. As unconditional models, LLMs have emerging abilities such as deep semantic understanding, humanoid reasoning, human sentiment analysis, etc., which can identify and analyze the motives and behavioral patterns of attackers more efficiently, providing a more comprehensive perspective on detecting textual content threats. From this part, we found that LLMs can improve the accuracy and generalization of textual threat detection compared with previous models. Meanwhile, some limiting issues in existing works are being further studied. All related works in this part are summarized in Table 4.

3.3. Malicious Code and Malware Detection

Code fragments in programs can be regarded as special strings or sequences that are restricted by strict syntax rules and internal logic. Quite a few studies or security products (e.g. Copilot, Deep Code) on code analysis have indicated that DL models can effectively extract hidden information from codes to be used for vulnerability mining or security issues analysis. In this view, some researchers used LLMs to detect and analyze malicious codes or malware. It should be noted that there is a certain difference between detection works and vulnerability mining. Vulnerability mining focuses on discovering and exploiting flaws or bugs in system programs, but threat detection for codes focuses on analyzing whether codes or programs are harmful. Although the detection of malicious codes and malware is highly correlated with comprehensive security scenarios, we still chose to discuss this topic separately. Because abundant literature

Table 4

Thematic works about LLMs on Textual Threat Detection and Social Engineering Prevention

Tasks	Effectiveness	Limitations	literatures
Threat Text Collection	Training DarkBERT on dark web ;Performed well in text generation.	Limited in non English tasks ;There is a strong dependence on domain data.	Jin et al. 2023
	Phishing emails by GPT-3 is slightly inferior to manually crafted ones.	Results may be limited to the content, type, and other factors in experiment.	Sharma et al. 2023
	SG-CSE BERT for analyzing historical information and predicting attack.	Despite of convenience to deploy, it has a small scale and average accuracy.	Tsinganos et al. 2023
	A framework for phishing attacks using existing conversational models and an instantiation testing process is provided.	Influenced seriously by prompt injection; the quality evaluation are not complete; Method has short-term effectiveness.	Shibli et al. 2024
Threat Text Detection	By combining Transformer and Ensemble learning with GPT-3, the accuracy of email classification reached 99.91%.	Data set only involves English messages; the scalability of the detection model is poor.	Ghourabi et al. 2023
	Using basic models such as GPT-4, the accuracy of phishing website detection tasks can reach up to 98%.	Results are unstable and susceptible to prompt or injection attack; The training data has a deadline time node.	Koide et al. 2023
	The accuracy, recall, and F1-score of threat and hate speech detection on social media are superior to traditional and DL models.	Language types for detecting have certain limitations; when applied to other languages, the model needs to be re-trained.	Bilal et al. 2023
	Effective in message content analysis, URL classification, etc.;Accurately identify phishing for mobile devices .	Not Mentioned	Kumarasiri et al. 2023
	Designing problems, LLMs can infer attributes , construct feature representations for downstream classifiers.	Not Mentioned	Nahmias et al. 2024

has shown that using LLMs for security analysis on codes is as appropriate as security analysis on natural language. This process is roughly divided into the main stages of code or software collection and processing, malicious code identification, malicious code clustering, etc. And LLMs are mainly applied in the code collection and static analysis of codes, as shown in Figure 4.

A.LLMs for Malicious Code and Malware Collection

It is highly effective in some adversarial tasks by generating additional datasets with specific features. LLMs are competent for generating high-quality code and tools, which can enhance the comprehensive detection capabilities of various models for malicious code or malware detection. Botacin 2023 explored the ability of ChatGPT to generate malware using GPT-3. Although They found that GPT-3 still struggles to generate complete malware samples from complete descriptions, it can easily construct malware through building block descriptions and generate multiple malware variants. And Charan et al. 2023 systematically generated executable codes for the top 10 MITRE techniques of 2022 using ChatGPT. Comparing its performance with Bard in generating malicious scripts, GPT was found to be more prone to misuse. Meanwhile, Pa Pa et al. 2023 used ChatGPT and Auto-GPT to generate seven malware programs and two attack tools, which indicated the possibility of generating functional malware within 90 minutes, including debugging

time. Through prompt engineering, ChatGPT created executable malicious codes, showing greater responsiveness and the ability to debug. Concurrently, a series of adversarial detection frameworks based on LLMs are also gaining popularity in the fields of automated offense and defense, as well as adaptive threat detection. Hu et al. 2021 explored DL-based malware detection and proposed a novel Adversarial Malware Generation (AMG) framework, named MalGPT. They used GPT to analyze malware byte sequences, achieving successful single-query black-box evasion models, and effectively circumventing malware detectors with just one query. MalGPT exhibited outstanding performance in handling eight major malware categories, with an evasion rate exceeding 24.51%, significantly outperforming existing benchmark methods. This study also incorporated malware source code into perspective analysis, anticipating that this multi-view deep learning approach could further enhance evasion capabilities. And Devadiga et al. 2023 proposed the GLEAM model, which combined hex code and opcode features with LLM embeddings to generate synthetic samples that are similar to evasive malware. By using LLM to create coherent and structured samples, they increased the average evasion rate by 25.0%, which demonstrated the ability of LLM to generate adversarial malware samples. Not only can these work aid in generating security test cases, and swiftly uncovering new threats, but they also hold significance for

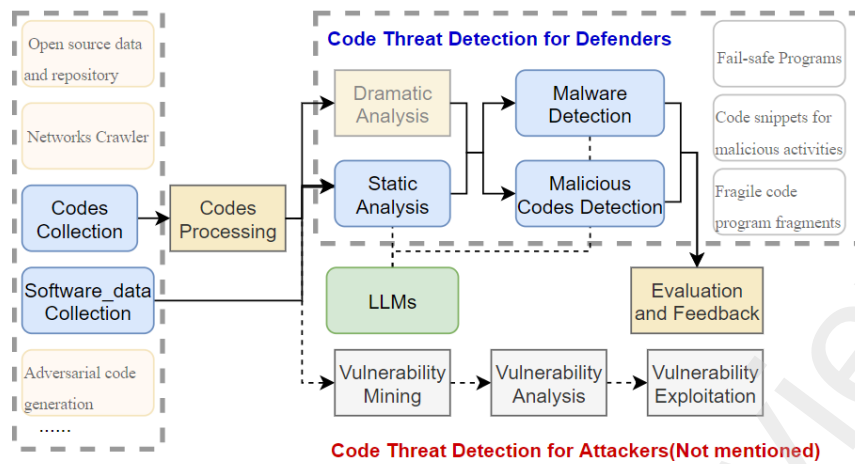


Figure 4: A diagram about LLMs in Malicious code and Malware detection

understanding and combating malware, especially in real-world detection with data imbalances.

B.LLMs for Malicious Code and Malware Detection

In addition to applying LLMs to the generation stage of malicious samples, some researchers prefer to use them in the static analysis or specific detection stage of malicious codes or malware. Demirci et al. 2022 used a combination model of Stacked BiLSTM, DSLM-GPT2, and GLM-GPT2 for malicious code s and malware detection, which is significantly superior to other DL models, especially in detecting malicious behavior in PE files. And Gao et al. 2022 extracted CFG from PE files and used large-scale pre-trained model MiniLM to generate node features of CFG. Then they employed the extracted CFG into feature vectors and used multi-layer perceptron for classification. Moreover, L. Li et al. 2023 used GPT-3.5 and ChatGPT for malicious webpage detection. Their experimental results show that its performance is comparable to or even better than DL methods, proving the feasibility of using LLM to detect network security threats. Next, Xiang Li et al. 2023 employed the Llama-7b model and image analysis of PE files for ransomware detection. By converting PE files into grayscale bitmap images, they can analyze them using the Llama-7b model. This method can achieve a significant true positive rate and minimal false positives and negatives and is effective for ransomware variants. Besides, García-Soto et al. 2022 detected malicious behavior in Android applications by using the CodeT5 pre-trained language model. They first generated embeddings from chunks of the source code after decompiling the Android application to Java, then used them for the BiLSTM classification model. As mentioned, all of these works attempted to combine LLMs with existing malicious codes and malware detection methods. Luckily, most of these results showed the feasibility of that.

In this part, LLMs can learn existing code fragments to build new security test samples, and migrate them to other detection scenarios, optimizing scalability, robustness, and generalization ability of malicious code and malware detection. By the way, most researchers have found that LLMs are

worthwhile for detecting and analyzing code and assessing program risks. However, it may be hard and unsuitable for LLM to counterattack threat code or fix vulnerabilities. The relevant work in this section is summarized in Table 5.

3.4. Intrusion and Threat Discovery in Comprehensive Security Scenarios

Furthermore, LLMs also have a wide range of applications in subtasks of IDS or threat discovery in practical security scenarios, including malicious traffic detection, security log analysis, potential malicious behaviors identification, etc. Although APT tends to be increasingly complex, covert, and diverse threat characteristics, LLMs possess advanced abilities of multi-modal information fusion and enhanced detection context. So we did not choose to list research work from the perspective of a single input data object but chose to analyze threat detection as a systematic work. This process mainly includes data collection and processing, threat feature extraction, correlation analysis and matching, and target scene threat recognition. LLMs nearly have traces of application in each stage, as shown in Figure 5.

A.LLMs for Security Data Collection

The scale, type, and complexity of data that needs to be detected vary in different security scenarios. Usually, it is necessary to extract and process a large amount of network traffic data from the detection object before detection, including various protocol packets, log files, and system event information. These data are not only massive in scale but also diverse in type, involving various security threats, ranging from simple malware propagation to APT. LLMs have unique advantages in processing, analyzing, and storing large-scale data, making them widely used in secure data collection. Setianto et al. 2021 developed GPT-2C, specifically for parsing dynamic logs generated by Cowrie SSH honeypots. This model was designed to overcome the limitations of traditional information transformation plugins when dealing with dynamic log topics such as user-generated content. The model was evaluated using a question-answering task, where the context consisted of a series of utilities and

Table 5
Thematic works about LLMs on Malicious Codes and Malware Detection

Tasks	Effectiveness	Limitations	literatures
Malicious Codes and Malware Collection	GPT-3 can construct malware via building block descriptions and generate multiple malware variants. ChatGPT can accelerate the generating of malicious code. ChatGPT generate functional malware (up to 400 lines) within 90 minutes. Significantly improves the chance of single-shot evasion. Effectively expanding the space for adversarial malware generation.	GPT-3 has trouble generating entire malware samoles from complete descriptions Reluctance in generating malicious code for corresponding. AugoGPT does not ease the hurdle of generating right prompts for malware. Excessive reliance on querying a malware detector multiple times. Not mentioned	Botacin 2023 Charan et al. 2023 Pa Pa et al. 2023 Hu et al. 2021 Devadiga et al. 2023
Malicious Codes and Malware Detection	Pre-trained models improved DSLM-GPT2 and GLM-GPT2 detection. Demonstrated 0.9977 in the Area Under Curve metric and achieved a 97.44% ACC when FPR was 0.1 %. Achive comparable or even better performance than deep learning baselines by analyzing webpage text content. High accuracy rates in identifying ransomware. Proved the feasibility of pre-trained models for code embeddings generation.	Data set only consisted of a few hundred malware and benign files. The process of extracting sample CFG is very time-consuming. ChatGPT sometimes refuses to answer by responding "Unable to determine, need more information". Employed data set omits older ransomware variants. Average accuracy score is 7% lower than the best result obtained in the origin paper.	Demirci et al. 2022 Gao et al. 2022 L. Li et al. 2023 Xiang Li et al. 2023 García-Soto et al. 2022

the answers were the utilities predicted by the model. It achieved an inference accuracy of 89% in new domains, with an acceptable execution latency. Data preprocessing and visualization for Cowrie instances, including key metrics like log volume and inference latency, were conducted using Elasticsearch and Kibana dashboards. The simplicity and rapid adaptability of GPT-2C indicate the potential application of LLMs in security raw data processing, Endpoint Detection and Response (EDR), and Security Information and Event Management (SIEM). And Kholgh et al. 2023 introduced an end-to-end PCA-GPT tool, leveraging GPT-3 to generate network traffic. This synthetic data can substitute real data in tasks like threat modeling, compensating for possible distribution shortcomings in actual datasets and thereby improving the generalization capability of detection. Besides, Deng et al. 2023 discovered that LLM excels in certain sub-tasks of network penetration testing, assisting manual efforts in comprehensive test scenario comprehension. Developing PENTESTGPT, an automatic penetration testing tool, they effectively minimized loss features by employing sub-modules to handle different sub-tasks. Assessments showed that PENTESTGPT's task completion rate surpassed GPT-3.5 by 228.6%, gaining over 4700 stars on GitHub. Then, PENTESTGPT has vast potential in security data generation and real-world attack-defense drills and could be further developed for automated offensive and defensive games to adapt to new threat forms and strategies. Then, Ohjuwon et al. (2023) invented a patent

utilizing LLMs for processing security raw data by inputting prompts into the application programming interface (API) of GPT, facilitating the preprocessing of raw security data. This application enabled the raw data to be transformed into a question-answer format based on the prompts, thus generating original and augmented datasets for fine-tuning models. This illustrates the key role of LLM in processing and transforming security raw data, particularly in parsing dynamic and complex logs. The advantage of this method lies in its ability to quickly generate data suitable for fine-tuning, saving the time and cost of creating datasets, and thus enhancing the efficiency of processing security data. In addition, there are still many other works involving LLMs simulating the generation of security data corresponding to different security issues in different security scenarios, which can provide more complete data supplementation for the training of detection models.

B.LLMs for Threat Feature Extraction and Analysis

Using LLMs for threat modeling, threat feature extraction and deep relationship extraction on the obtained security data set can effectively compensate for the shortcomings of traditional threat detection models in this regard. Akbar et al. 2022 utilized the deep semantic learning capacity of the RoBERTa model to infer semantic associations between descriptions of attack techniques and defensive countermeasures. The results indicated that this method could unearth some hidden relationships within security

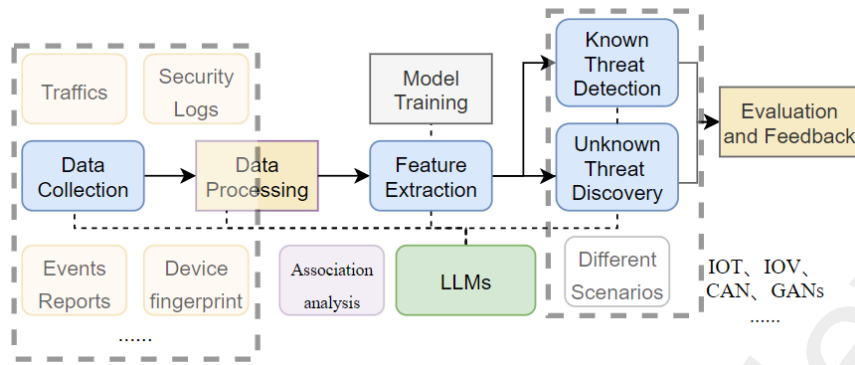


Figure 5: A diagram about LLMs for Intrusion and Threat Discovery in Comprehensive Security Scenarios

data, providing useful inferences and suggestions for security analysis and threat detection with commendable precision, especially in comparison to baseline methods like word2vec and TFIDF. And Alam et al. 2022 proposed the LADDER for extracting attack patterns from CTI to optimize threat detection. The LADDER can effectively mine complex information within CTI reports using RoBERTa, identifying key features of existing and emerging threats, which can assist security analysts and detection systems in preemptively recognizing potential attack methods for proactive defense. They also contributed a new open-access malware data set for training future CTI models. Besides, Shi Z. et al. (2023) combined BERT and LSTM to develop a fixed model for classifying malicious traffic. This model can effectively integrate global features captured by BERT with time-series features captured by LSTM, enhancing the precision of malicious traffic identification by matching. Upon the USTC-TFC data set, this method has improved the accuracy of traffic classification to 99.49%, a 0.2% increase over baseline models. As for threat modeling, Gadyatskaya et al. 2023 utilized GPT-3.5 to automatically create attack trees, a universal method of threat modeling representing network threats from an attacker's perspective. Their work demonstrated that GPT-3.5 contains sufficient information about cyber threat and their relationships to generate attack trees, improving the automation level of threat detection and discovery of unknown threats. However, limitations like an inability to infer detailed operators were noted. Moreover, Boffa et al. 2024 designed the LogPrécis model based on various models including GPT-3 Davinci (175B), BERT, CodeBERT, and CodeBERTa, for automated analysis of text-based Unix shell attack logs. Utilizing the MITRE ATT&CK framework as a guiding structure to capture the causes of attacks, this method automatically learned the attacker's strategy by processing original shell sessions, thereby reducing the number of attack logs and enhancing attack analysis support. Meanwhile, Lin et al. 2022 proposed a pre-training framework ET-BERT for encrypted traffic classification. This framework utilizes large-scale unlabeled encrypted traffic to learn a universal datagram representation for a series of encrypted traffic classification tasks. ET-BERT was tested in two traffic-specific self-supervised pre-training

tasks, such as the masked BURST model and homologous BURST prediction, which capture byte level and BURST level contextual relationships to obtain a universal datagram representation. ET-BERT has strong generalization ability and has achieved new and state-of-the-art performance in five encryption traffic classification tasks, including general encryption application classification, encryption malware classification, VPN encryption traffic classification, Tor encryption application classification, and TLS 1.3 encryption application classification. It is significantly better than existing works by 5.4%, 0.2%, 5.2%, 4.4%, and 10.0%. Moreover, Meng et al. 2023 proposed a pre-trained traffic understanding and generation model, NetGPT. By shuffling header fields, segmenting packets in the flow, and introducing various task labels, the training efficiency and effectiveness of downstream tasks such as traffic classification, attack detection, resource scheduling, protocol analysis, and traffic generation have been further optimized. NetGPT also addressed the heterogeneity and diversity of network traffic, as well as the different dependency relationships of downstream network tasks. Last but not least, through multimodal aggregation and collaborative technology frameworks of large models and small models, LLMs can demonstrate unique advantages in threat detection and comprehensive analysis, which are being introduced in some latest work.

C.LLMs in Real-world Threat Detection Scenarios

In addition to their contributions to security data and threat feature engineering, LLMs can be integrated into actual threat detectors in different network environments to optimize their output. Wang et al. 2024 proposed a lightweight IoT intrusion detection model named BT-TPF based on BERT-of-Theseus. It reduces feature dimensions using Siamese networks and employs a large visual transformer as a guide model to teach the smaller Poolformer model. After knowledge distillation, the BT-TPF model's parameter count was reduced by 90%, requiring only 788 parameters. On datasets such as CIC-IDS2017 and TON_IoT, the model achieved an accuracy rate exceeding 99% in traffic-based intrusion detection, significantly outperforming traditional deep learning methods and the latest models, making it particularly suitable for resource-constrained IoT. And Xuemei Li et al. 2023 developed two new models,

Table 6

Thematic works about LLMs on Intrusion and Threat Discovery in Comprehensive Security Scenarios

Tasks	Effectiveness	Limitations	literatures
Security Data Collection	Using GPT-2C to parse dynamic logs;Being suitable for EDR and SIEM.	Not Mentioned	Setianto et al. 2021
	Using GPT-3 to generate composite data of network traffics;compensate for the distribution defects of the data set.	Only two protocols were considered;training data and training duration are limited;CLI.	Kholgh et al. 2023
	Generate security examples ; Assist in automated penetration, adapting to new forms and strategies of threats.	Not Mentioned	Deng et al. 2023
	Convert original security data into Q&A format; Being suitable for parsing dynamic and complex logs.	Not Mentioned	Ohjuwon et al. 2023 (patent)
Threat Feature Extraction and Analysis	Infer semantic relationship between attack description and defense counter-measures.	Attack and defense information contained in the corpus is limited;Indirectly applied to threat analysis.	Akbar et al. 2022
	Using LADDER to detect from CTI for attack pattern extraction and trend analysis,threat hunting and prediction.	Only verified in Android and enterprise networks; Test samples is limited.	Alam et al. 2022
	Using BERT-LSTM to improve accuracy of malicious traffic classification, combing global series characteristics.	Do not take into account malware traffic on mobile phones.	Shi et al. 2023
	Using GPT-3.5 to generate attack tree automatically, the automation of threat detection is improved.	Can not reason with refinement operators;Hallucinations of LLMs;lack of checking integrity of attack tree.	Gadyatskaya et al. 2023
	LogPrécis based on multiple models is used to enhance UNIX shell attack log analysis and feature extraction.	LogPrécis fingerprint may be the same even for different attacks. This design can lead to misclassification.	Boffa et al. 2024
	ET-BERT for encrypted traffic classification and malicious traffic feature extraction.	Insufficient innovation and modification for BERT.	Lin et al. 2022
Real-world threat detection scenarios	NetGPT for traffic analysis and generation, optimizing downstream tasks.	no benchmarks and the compared downstream tasks are limited,.	Meng et al. 2023
	Accuracy of intrusion detection using BT-TPF in IOT is over 99%, suitable for resource constrained situations.	Not Mentioned	Wang et al. 2024
	CAN-SecureBERT and CAN-LLAMA2 for intrusion detection applied in CAN significantly reduce FPR.	LLMS' reasoning speed is limited by computing resources.	Xuemei Li et al. 2023
	HuntGPT anomaly detection system based on ML and XAI. Provide interpretable threat representation.	Achieve lower success in fundamental cyberspace security concepts.	Ali et al. 2023
	Using SecurityLLM for common threat detection and incident response.	Further fine-tuning is required in specific tasks;continuous updates are required to maintain effectiveness.	Ferrag et al. 2023

CAN-SecureBERT and CAN-LLAMA2, for intrusion detection in Controller Area Networks (CAN). Notably, the CAN-LLAMA2 model reached a balanced accuracy rate (BA) of 0.999993, a precision (PREC) rate, a detection rate (DR), and an F1 score, with a false alarm rate (FAR) of only $3.1e-6$, which is $1/52$ of the FAR of the most advanced model MTH-IDS. Experiments demonstrated that transformer-based models and LLM have outstanding performance in CAN intrusion detection, particularly in reducing FPR.

Additionally, Ali et al. 2023 designed HuntGPT, a network anomaly detection system combining ML and explainable AI (XAI), based on GPT-3.5-Turbo and random forest classifiers. On the KDD99 data set, HuntGPT provided an intuitive user interface and comprehensible threat representation by XAI frameworks like SHAP and Lime. The system demonstrated significant abilities in cybersecurity knowledge, achieving success rates between 72% to 82.5% in standardized cybersecurity certification exams, including

those assessed by Certified Information Security Managers (CISM). This blend of LLM and XAI in a conversational agent effectively enhances the operability and interpretability of the intrusion detection system. The approach offers a program of interpretable, interactive threat detection, reducing manual intervention and improving the efficiency of threat detection and presentation. As for some universal threat detection based on LLMs, Ferrag et al. 2023 have successfully fine-tuned SecurityBERT and FalconLLM, leading to the creation of SecurityLLM, a domain-specific LLM for threat detection. In this fixed model, SecurityBERT serves as a threat detection mechanism, while FalconLLM aids in event response and recovery. This approach has shown superior performance in terms of precision and recall on test datasets compared to most previous methodologies. Besides, this model can identify 14 different types of threats with an overall accuracy of 98%, using various security datasets. However, it requires continuous updates on the latest real datasets to maintain effectiveness.

As is shown above, LLMs are rarely applied independently for threat detection in most cases, but LLMs can be nested into existing threat detection frameworks for optimization in a certain stage. In this way, not only do LLMs maintain excellent performance on conventional detection indicators, but there is also optimization potential in terms of automation, robustness, and generalization ability of threat detection. But not all tasks in different threat detection scenarios, LLMs are better or appropriate. There are still lots of issues being studied. The relevant work in this part is summarized in Table 6.

4. Review Methodology

According to the description by Snyder 2019 and other similar literature review methodologies, our reviewing works mainly consist of three phases: (i) review designing, (ii) review conducting, and (iii) result analysis. Each phase was chronologically performed in detail as follows.

4.1. Review Designing

To begin with, our reviewing works were guided by review rules and research questions.

4.1.1. Review Planing

The framework established for investigation delineates a sequence of actions for undertaking the systematic review and meticulously choosing primary references.

I.Scope of study

Since we attempt to explore how to apply LLMs in different scenarios of cyber threat detection, we can not only discuss the direct and independent use of LLMs for threat detection, but also research how LLMs are combined with other models. Meanwhile, inclusion and exclusion criteria should be verified according to Table 7.

II.Search Engines and Databases To systematically collect relevant literature, this study primarily utilized the following five databases: ScienceDirect, Web of Science, SpringerLink, IEEE Xplore, and DBLP. Additionally, search

Table 7

Inclusion and exclusion criteria

Inclusion Criteria	Exclusion Criteria
Literature accessed from superior journals or conferences.	Literature originates from unofficial or unknown source.
Literature has high correspondence with theme topics.	LLMs were not applied for cyber threat detection.
Experiment is clear and convenient to reproduce.	Experiment relatively unclear or hard to reproduce.
Published in latest three years.	Published too long ago.

engines such as Google Scholar and Microsoft Academic were employed to supplement the review material.

III.Key Words and Query Statement Usually, the search strategy aims to identify all works associated with several primary search terms, incorporating synonyms, abbreviations, and various spelling variations for each term. According to theme, LLMs and cyber threat detection are two key topics. Some keywords associated with LLMs include large language models, GPT, Bert, llama, etc. Words associated with cyber threat detection include malicious traffic, security log analysis, malware classification, etc.

For example, we have preliminarily constructed some retrieval statements based on ((Large language model)OR(LLM) OR(GPT)OR(Bert))AND((Cyber threat detection)OR (malware detection)OR(networks traffics)), and sorted all results in each database according to relevance.

4.1.2. Research Questions

To learn how to apply LLMs in cyber threat detection better, the appropriate and effective selection of research questions is necessary. In this light, four research questions were defined as follows, which were answered in Section 4.3.

RQ1. Did LLMs perform better than traditional models in the specific task mentioned? The essence of this question is to explore which tasks LLMs are suitable, focusing on which tasks LLMs improved performance and which part LLMs are unsatisfactory.

RQ2. What methodologies were used in each study? We must be concerned about the convenience of relevant works in actual scenarios. Because the effectiveness of LLMs has a relatively harsh demand for data and computing resource environments.

RQ3. What were any limitations that existed in this study? It aims to find if there exists serious prejudice to support for LLMs better or worse.

RQ4. What were the major findings in this study? Apart from the advantages or weaknesses of LLMs in each task, further analysis of the authors' discussion may help us apply LLMs in security products better.

4.2. Review Conducting

The whole process of our reviewing works have been described in Fig 6.

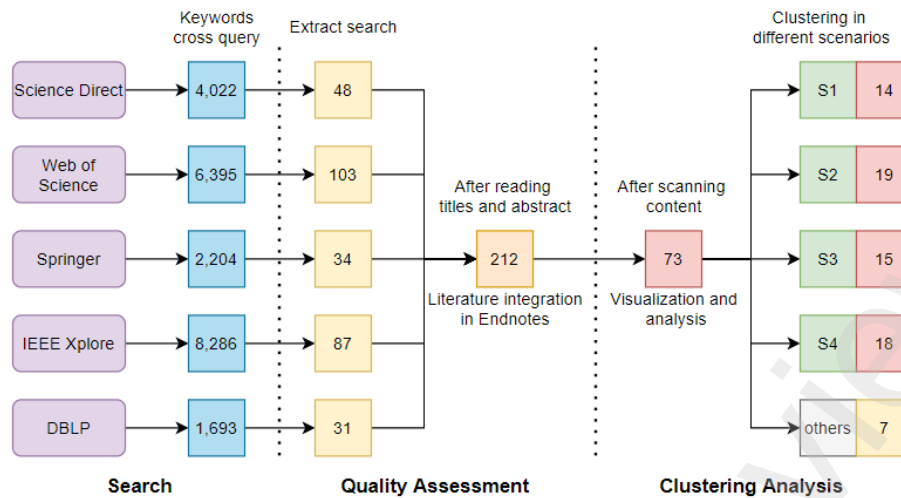


Figure 6: The process of reviewing works

As shown above, 212 works of high relevance were gradually selected from the initial works after query and screening work. And we verified thematic works by scanning content in EndNote, a literature management tool. After evaluating the impact, timeliness, and application value of each study, a total of 73 works were found to meet the inclusion criteria. Most of these works can be clustered into different threat detection according to manual analysis based on results produced by VOSviewer, as shown in Fig 7.

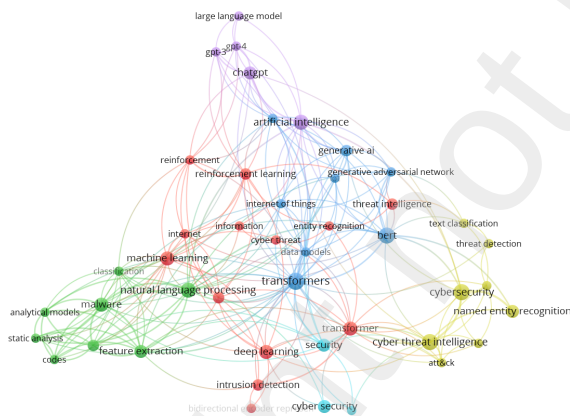


Figure 7: keywords clustering in selected thematic works

4.3. Result Analysis

Upon investigation of thematic works, we found significant merit in applying LLMs for threat detection, especially in tasks like heterogeneous raw data generation, detection on natural languages or other sequences (e.g. phishing SMS, threat speech, codes analysis), structured reports production, etc. Now we can answer questions mentioned in section 4.1.2.

I. LLMs are not always a better choice.

Not all threat detection and discovery works can LLMs get better performances than traditional methods. Some representative assessments of LLMs across different tasks are summarized in Table 8.

From Table 8, we can learn that there exist different voices about the effectiveness of applying LLMs in different tasks. Specifically, tasks involving ER, RE, structured information or sequence process, data collection for adversarial detection, etc. tend to benefit more from LLMs. This is primarily attributed to the proficiency of LLMs in understanding and generating natural language or serialized information. Additionally, LLMs can effectively analyze massive unstructured textual data, discovering hidden patterns or relationships that might be missed by traditional models. Besides, LLM is ideal for enhancing user interfaces and communications in security systems, thus improving the overall efficiency and user experience in threat detection. However, we can notice that the performance of LLMs is unstable due to some factors such as unresolved hallucination problems, the quality of datasets for retraining, and domain-specific knowledge for LLMs. In contrast, traditional models may be more suitable for applying directly in real-time intrusion detection, threat monitoring, and low-level threat analysis, due to their lower latency and more focused approach. Usually, these models are tailored to process specific types of data and threats, rendering them more effective in real-time or single-task detection and response. Then, traditional models can be more transparent and easier to interpret in terms of their decision-making process, which is essential in security contexts. Nevertheless, in a complex cyberspace environment, these methods, even the most advanced DL methods, can not detect threats effectively. In instances where the property is of high value, LLMs may become important even necessary. Although these findings may appear inconsistent with those of some researchers, they are not contradictory. Whether LLMs are better or not, it is generally inappropriate to separately employ LLMs for cyber threat detection. Most thematic works indicated it indubitable a better choice to

Table 8
Different assessment of LLMs in different tasks

Tasks	Literature	Assessment Based on Authors' Perspectives
Automatic Detection Reports	Perrina et al. 2023	High value in RE, CTI, and security reports production.
Data Processing for Detection	Kholgh et al. 2023	Convenient Convenient in generating traffic and other raw data.
SMS Spam Detection	Ghourabi et al. 2023	High ACC in SMS spam detection, but poor scalability.
CAN Instruction Detection	Xuemei Li et al. 2023	High ACC and low FPR, but unstable effectiveness.
Threat Modeling and Analysis	Gadyatskaya et al. 2023	Helpful, automation is improved but some problems need solving like hallucinations of LLMs.
Malicious Codes Collection	Charan et al. 2023	Helpful, accelerate generating malicious codes, but sometimes reluctant in corresponding.
Vulnerability analysis in CTI	Liu et al. 2023	Limited, ChatGPT still can't be used directly for VDM tasks.

integrate different methods like using LLMs to necessary stages to optimize a certain aspect of the detection scheme.

II. LLMs are mostly applied indirectly in research.

According to Section 3, it's an indispensable and more popular option of integrative methods, combining the strengths of both LLMs and traditional models. Most of the existing research was conducted jointly with GPT, BERT, and ML, DL methods. However, a segment of researchers have still innovated by developing their domain-specific LLMs for threat detection or other tasks. While GPT series LLMs were applied more in generating various data of traffic, codes, test samples, or other data, BERT series LLMs were more adapted in analysis works like RE and ER. Simultaneously, we found the domain-specific LLMs fine-tuned on GPT-2 or other previous open-source LLMs, are more favored. The reason could be there exist restrictions to access or retrain GPT-3 or later closed-source LLMs. Besides, some researchers have critically assessed the advantages and disadvantages of LLMs by comparative experiments. Utilizing LLMs directly for threat detection may be suitable only on a few occasions like phishing E-mail detection and codes static analysis etc.

III. there are still some limitations in this research.

Unfortunately, not all studies articulated fully detailed methods of how they develop models or shared datasets for readers to retrain local LLMs. We know the actual effectiveness of LLMs can be influenced by datasets and knowledge dramatically. Next, there are too few datasets involved in most comparative experiments, so we can not confirm whether LLMs can truly maintain the same performance on the vast majority of datasets. To verify the effectiveness of LLMs, more research needs to be conducted. Last, nearly all researchers have attempted to demonstrate the feasibility of LLM in specific tasks rather than multi-task scenarios. Scenarios in which LLMs and small models work together for threat detection have not been discussed in more detail. Consequently, there is a lack of unified standards or datasets to guide the implementation of LLMs in actual security systems.

IV. The LLM methods are considerable to optimize.

Although LLMs for threat detection are not perfect now, the vast majority of research indicates that LLMs have enormous potential in cyberspace threat detection and discovery, but there are lots of optimizations that need doing, especially how to provide high-quality retraining data for LLMs, how to make LLMs' performance more stable, how to decrease the cost of LLMs in real-world systems. Thus, more and more researchers are joining this field to explore a better solution or technical framework with LLMs for threat detection and discovery, especially cooperation between LLMs and traditional models.

5. Discussion

A. Summary

In this study, we found LLMs were applied in quite a few tasks of cyber threat detection and discovery, summarizing four security scenarios highly related to that. Thematic research has shown that LLMs have high potential in tasks such as threat intelligence processing, natural language threat detection, and static code detection, which is consistent with the conclusions of some mainstream technology reports. Some highly relevant thematic work with comprehensive analysis may help security workers build domain-specific LLMs for cyber threat detection in specific scenarios.

The innovation of this review lies in its approach to addressing current problems in threat detection, retrospectively tracing the improvements made by LLMs in different specific security scenarios, rather than sequentially listing these works categorized by types of input data. The main reason is that we found not all aspects of threat detection benefit from LLMs. Meanwhile, it does not effectively highlight where LLMs are most valuable to discuss the combination of all threat detection tasks with LLMs.

However, there still exist some limitations in the review. For example, all thematic works presented are mainly sourced from literature databases within three years, without further expanding the scope to include more patents, products, books, white papers, and organizational reports. Additionally, the organization of the review is more focused on

the benefits of the logic perspective, lacking more detailed analysis and related works on specific threat detection tasks, such as traffic detection and log analysis.

B. More Topics

Besides, it's worth talking more related issues related to our topics, most of which were also discussed by security researchers these days.

I. The contradiction between LLMs and threats.

LLM, one of the most advanced AI technology tools, is a double-edged sword for cyberspace security. It's argued that more threats can be disposed of by LLM while more threats are generated by LLM. Specifically, LLM reduces the threshold for hackers and the process of attack-chain faster without restrictions. In contrast, LLM can provide a series of supports to address these new problems and shorten the response chain of threats.

II. The relationship between LLM and cyber security.

There exists a complementary relationship between the LLM and cyberspace security. On one hand, security for LLM, which is dedicated to securing LLMs, is an important prerequisite for achieving secure applications of LLM. On the other hand, LLM for security, which is dedicated to using LLM to solve security problems, can promote the resolution of LLM's security.

III. which one is better, efficiency or performance?

In the practical optimization of LLM, it is a pivotal dilemma whether a larger LLM is inherently better or efficiency holds greater value. Generally, larger LLMs imply advanced capabilities and improved effectiveness, yet concurrently incur higher temporal expenditure as well as costs for training and operation. Conversely, heightened efficiency allows LLMs to respond more swiftly to tasks, striving to approach the upper limit of performance in their current size as closely as possible. Optimizing for both size and efficiency simultaneously is often not feasible, necessitating a tailored approach that aligns with the specific requirements of the detection tasks at hand.

IV. Is LLM for security trustworthy?

It was reported that users and vendors may encounter issues such as distrust and conflicts of interest (Gupta et al. 2023). The "Q&A alignment problem" refers to the situation where users can not always get what they want due to the restriction of output. And some countries even think LLM from other regions may threaten their security or help the locals break laws, and have strictly prohibited products based on LLM. So, is it trustworthy to use LLM to detect various threats without any side effects?

All of these similar problems are being discussed these days in different articles, reports, and even newspapers. It is not only a technical issue of how to integrate LLMs, threat detection, and cyberspace security more efficiently.

C. Future Works

Based on existing works, we think future research could focus on these areas. Based on existing research, future studies in this domain are recommended to explore the following areas:

I. Improvements in existing problems

It's a considerable option to study applications of LLM on security data processing, detection of linguistic threat carriers and serialized information, threat pattern recognition, adversarial detection, and structured report generation, as well as attempts to combine LLM with superior DL frameworks.

II. Exploring potential research directions

It's also worth investigating LLM in internal threat and espionage detection, acceptable security detection reasoning frameworks, threat modeling and prediction, and comprehensive analysis of various threat carriers simultaneously, such as traffic, logs, software, etc. Besides, We may recognize that LLMs are capable of not only identifying threats but also assisting in their control. This capability could advance cybersecurity situational awareness to a higher level of understanding, potentially introducing new definitions and entirely new technological frameworks for cyber threat detection.

III. More Interdisciplinary Crossover

Given LLMs' strengths in deep language understanding and human-like expression, we can integrate LLMs with cyber cognitive threat detection, hackers' psychology, and game theory. All of these works may enhance the breadth and effectiveness of threat detection indirectly.

IV. Challenges of Domain-Specific LLMs

It's necessary to figure out the security issues, ethical compliance, and privacy protection challenges inherent in LLM. Additionally, it is also a joint area of interest for both security practitioners and AI researchers to improve the interpretability of LLMs in threat detection.

We plan to delve deeper into some areas above, conducting research works to find more applications of LLMs on threat detection.

6. Conclusion

In this review, we have systematically explored a brief history of LLMs and cyberspace threat detection, analyzing the potential of LLMs in addressing existing challenges in threat detection. By investigating relevant literature over the past three years, we discussed applications of LLMs in different tasks of typical security scenarios, enhancing the adaptability and automation level of threat detection. In the context of this, we summarized the current development status in this interdisciplinary field and explored the potential works and related topics of further LLM applications. This review focuses on providing a comprehensive perspective on the use of LLMs in threat detection, offering valuable insights for researchers interested in designing and optimizing threat detection using LLMs.

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